

Raghav Senthil Kumar

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, Agentic AI, BuildIllinois, Innovation LLC, Agentic AI

EXPERIENCE

Founding Software Engineer | Clusion

September 2025 – January 2026

- Built the document search engine (*Python FastAPI*) for a competitive high school debate research SaaS platform hosted on *AWS*, developed by an *Agile* team of **10 engineers** and drawing **10K+ site visitors**
- Indexed **600K+ debate-evidence documents** on *Elasticsearch* via *FastAPI*, returning results in **under 1 second**
- Designed and built a text-retrieval algorithm that extracts evidence segments from debate cases by validating source URLs and filtering statistical outliers with **IQR** on the word-count distribution of tag, citation and evidence text
- Developed a de-duplication algorithm that flagged **30%** of evidence documents as identical, reducing index bloat
- Implemented recency-based relevance so evidence from the current month's debate topic ranks first in results

Product Development Intern | Jetson

January 2025 – February 2025

- Conducted **7 iterations of UI/UX testing** across 3 experimental features on the Jetson App (100K+ users)
- Delivered a **5-page report** recommending strategies to optimize user engagement directly to Jetson's CEO
- Redesigned user onboarding on *Figma*, cutting it from **12 to 5 pages**, validated via **A/B testing** on 30 test users

Data Analytics Intern | STEM E

June 2024 – July 2024

- Analyzed 3 years of TikTok data (**200+ posts**) with *Python (pandas)*, *Hootsuite*, *Power BI* and *Microsoft Excel*
- Identified that "science fun facts" posts comprised 69.3% of views, despite accounting for 7% of all content
- Pitched a "fun facts" strategy to the Executive Director, who adopted it; engagement rose **24% (8K+ views)**

PROJECTS

Truck Pricing VLM | [Live Website](#) | [Source Code](#)

September 2026

- Led team of **4** to build a pricing algorithm for **Kamion (YC S22)**. **2nd of 7 at UIUC Founder's Hackathon**
- Scraped **15,000** truck-listing images for *data mining*, auditing quality via de-duplication and non-USD filtering
- Designed a *JSON schema* for a **Claude Sonnet 4.6 agent** to extract condition, brand, model and year from photos
- Revised the schema after each wrong prediction until **10 consecutive images** were correct on each of **3 features**
- Built a *quantile regression* pricing model on the extracted features, reaching **87% accuracy** on **30 samples**

SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Engineered a **suite of 3 edge AI tools** that collects data regarding a school's energy, water and food consumption
- Built a *Python FastAPI* backend that ingests live water, waste & energy data from all 3 tools, runs **4 Random Forest models** to detect anomalies & deliver actionable steps via **Gemma Voice LLM** on a *Next.js* dashboard
- Led a team of 5 across project ideation, development and presentation. **Finalist at USAII Global Hackathon**

Citation Chain | [Live Website](#) | [Source Code](#)

June 2026

- Built an interactive citation graph (*React Flow*) that maps a paper's outbound citations across **26 research fields**
- Built a dual-API ingestion pipeline resolving papers via *OpenAlex* & backfilling metadata from *Semantic Scholar*
- Deployed a **RAG agent (Claude Sonnet 4.5)** that retrieves session-scoped vector documents (*Supernemory*)
- Stored paper data on *Supabase (PostgreSQL)* and wrote *SQL* queries to filter papers by field, year and citation count

EcoBin | Research Paper | [Source Code](#)

March 2026 – May 2026

- Designed a two-stage *EfficientNetV2-S* neural network (**4.39M** parameters) in *PyTorch*, trained on **15K+ images** to classify waste into 4 disposal pathways at **96.13% accuracy**. Published a paper on *arXiv* and 2 datasets on *Kaggle*
- Built a synthetic data generation pipeline using *U2-Net* segmentation and alpha-compositing to generate 9K images to train the model to detect contamination in recyclable waste. **Placed 2nd at the Arizona State Science Fair**
- Engineered a smart bin prototype using *Arduino Uno* that sorts the waste item with **less than 2 second latency**

Chronos | [Live Website](#) | [Source Code](#)

December 2025

- Built a **scheduling agent (React/Firebase)** that auto-books meetings for large groups (30+ ppl) on Google Calendar

- Designed a **5-step *Python* scheduling algorithm** that scans calendar availability, returns a ranked list of time slots and sends email confirmation for each booking. Validated algorithm across **14 unit and integration test cases**
- Created a lobby system that enables the host to authenticate invited participants and queue follow-up meetings
- Licensed Chronos to **8 startups** & signed **2 LOIs worth \$3.5K**. **Won 2025 Congressional App Challenge**

TECHNICAL SKILLS

Languages: Java, Python, C++, JavaScript, SQL, HTML/CSS
Frameworks & Libraries: FastAPI, Flask, Next.js, React, React Flow, Node.js
Databases & Search: Elasticsearch, PostgreSQL, MySQL, Firebase, Supabase
Cloud & DevOps: AWS, Google Cloud, Vercel, Docker, Git, GitHub, CI/CD
AI/ML: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, Grad-CAM, Hugging Face, Claude API, Gemma 4, Vercel AI SDK, Supermemory, RAG
Data & Analytics: pandas, NumPy, Power BI, Microsoft Excel, Hootsuite, A/B Testing
Tools & Practices: Agile, API Development, Google Calendar API, OpenAlex API, Semantic Scholar API, Figma, Kaggle, Arduino IDE, Bun, VS Code, Claude Code, Codex
Concepts: Data Pipelines, Data Ingestion, ETL, Data Quality, Data Validation, Data Cleaning, Data Mining, Data Visualization, Statistical Analysis, Prompt Engineering, AI Agents, LLMs, Generative AI, REST APIs, Full-Stack Development, Unit Testing, Authentication

CERTIFICATIONS

Harvard CS50x: Introduction to Computer Science	November 2024
HackerRank Problem Solving (Basic) Certificate	August 2026